

Fusing Noisy and Corrupted MRI Images

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Final Report

EE4C03 Statistical Digital Signal Processing and Modelling
Delft University of Technology

November 2, 2023

1 Introduction of the Problem

Magnetic Resonance Imaging (MRI) is an imaging technique that creates highly detailed pictures of the body's structures, including soft tissues, nerves and blood vessels. When compared to other imaging techniques, MRI provides a great soft tissue contrast. Therefore, it is one of the most important imaging techniques [1]. However, the quality of MRI images can be compromised due to factors such as patient motion and noise. In this project, we aim to address this problem by using tools such as optimal filtering and spectral estimation.

2 Motivation

The quality of MRI images is important because it effects the accuracy of diagnosis and treatment. Therefore, enhancing image quality and reducing artifacts that are caused by noise or motion is very important. By doing so, healthcare professionals can provide their patients with more reliable information leading to better outcomes when the health of the patient is considered.

3 Proposed Methodology

To create the MRI images, we will be implementing three main processes in our project. These are slice selection, k-space acquisition, and image reconstruction. Each process will be described in detail.

Slice Selection: In MRI, tomographic slices of the body or a target organ. To create tomographic slices, slice selection process is used and the target organ or part of the body that we want to image is excited. To achieve this, gradient fields are used and as a result spatial position becomes encoded in the received signal. [2], [3]

K-Space Acquisition: K-Space is a mathematical representation that captures spatial frequency information of the signal. When an MRI scan is performed, data is collected in K-Space before it's transformed into an image. In MRI, it is important to encode spatial information of the signal in the effective spin density with frequency-encoding. Effective spin density depends on varying number of scanning parameters such as echo time and repetition time; therefore, varying images are created. [2], [3]

Image Reconstruction: To reconstruct the images from the received MRI signals, we employ the inverse Fourier transform. [2], [3]

For the “good data” images without outliers from the datasets, fusion operation will be carried out, noise variance of the images will be estimated and the optimal way of fusing images that are coming from three channels will be discussed. To have good quality images from “bad data” images with outliers from the datasets, various techniques like smoothing, low-pass filtering, interpolation can be used. Also, images that are coming from different channels can be fused together. The order of fusion and reconstruction with the focus of eliminating the outliers will be discussed.

3.1 Explanation of Datasets

The datasets that we are given for this projects are MRI images of the human eye. Since patients are likely to blink during an MRI scan, the spectral lines get corrupted when the patient blinks. We have eight slice datasets, each of them having signals coming from three channels. Also there are two sets of data for each of the eight slice and channel datasets; as being “GoodData” and “BadData”. Below the spectral frequency observations of the k-space data of Slice 1 for both the “GoodData” and “BadData” can be found:

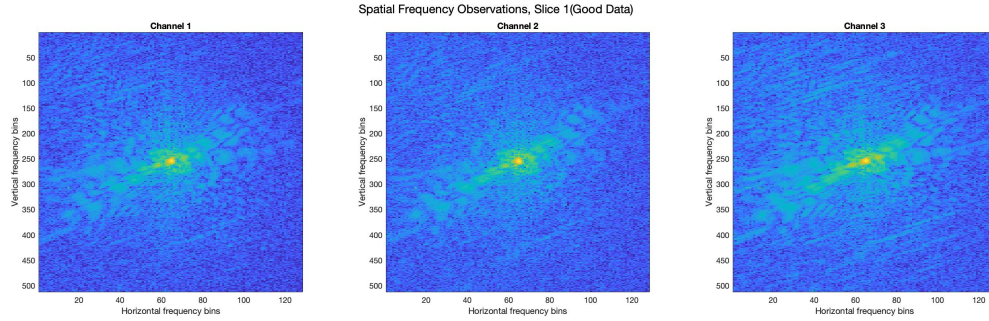


Figure 1: Spatial Frequency Observations, Slice 1 (Good Data).

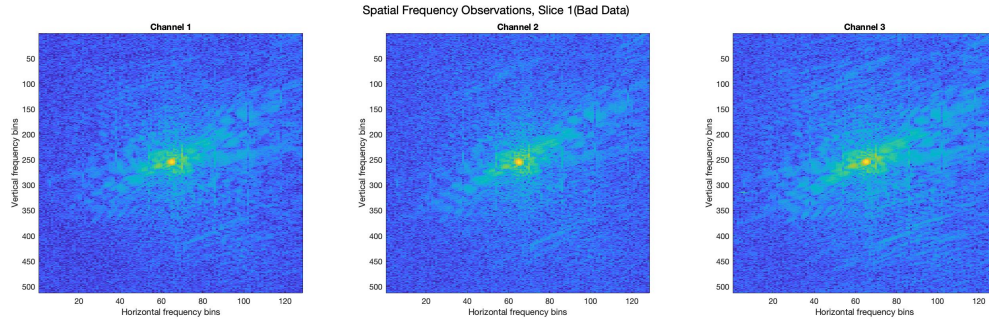


Figure 2: Spatial Frequency Observations, Slice 1 (Bad Data).

When Figure 1 and Figure 2 are compared, it can be seen that there are corrupted vertical lines in Figure 2. Figure 1 looks cleaner. The main objective of this project is to restore the corrupted vertical lines in the “Bad Data” images so that a cleaned image that looks like the “Good Data” images can be reconstructed.

3.2 Analysis Procedures and Models

The analysis procedures and models that are used in our project is based on the given datasets. The algorithms that are used relate to two important tasks, detection of the outliers and rejection of the outliers.

3.2.1 Detection of the Outliers

For the detection of outliers, two different methods were employed.

Using “Bad Data” Only

Firstly, “bad data” images were used only. By calculating the power of all the 512 columns in the “bad data” in k-space we were able to find the power spectral density of the images. This was done for each channel because outliers occur sparsely across the channels. The outliers presented themselves as local max values. The detected outliers with this method is presented below:

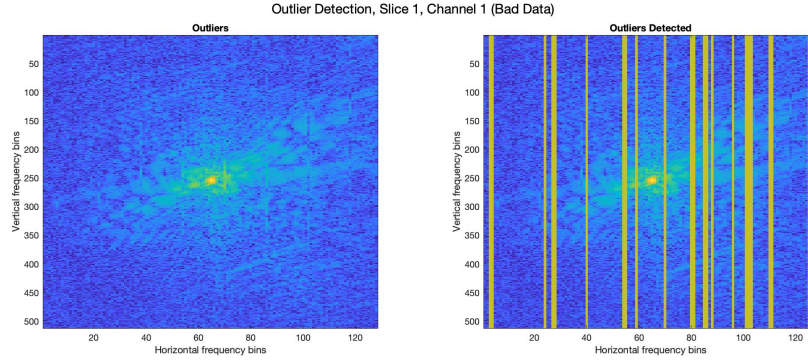


Figure 3: Outlier Detection, Using “Bad Data” Only.

As it can be seen, this method is able to detect outliers but it also classifies non-corrupted lines with high energy as outliers.

Using “Good Data” and “Bad Data” Together

To detect the outliers better, we used “Good Data” images and together with the “Bad Data” images. We found the difference between “Good Data” and “Bad Data” in each channel and then we calculated the power of all the 512 columns in the difference data. The outliers presented themselves as local max values. The detected outliers with this method is presented below:

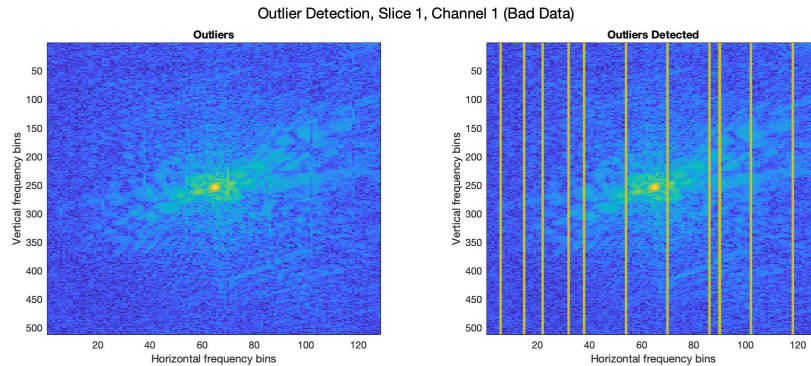


Figure 4: Using “Good Data” and “Bad Data” Together.

As expected, this method is able to detect outliers more accurately. It even detects the small vertical corrupted lines in the k-space data that is only visible to us when zoomed in.

3.2.2 Rejection of the Outliers

After outlier columns have been found via the technique explained above, several methods are applied to remove them.

Linear Interpolation

The linear interpolation method is simply based on linearly filling the corrupted columns using neighboring un-distorted columns. The average of the two neighboring columns of corrupted column is replaced with the corrupted column. The linear interpolation method is performed on the raw k-space data. Below, the formula used can be seen:

$$h(i, j) = \frac{1}{2}(h(i, j - 1) + h(i, j + 1)) \quad (1)$$

“ $h(i, j)$ ” stands for the row i and column j of the raw k-space data matrix, which is corrupted. “ $h(i, j - 1)$ ” and “ $h(i, j + 1)$ ” are the entries on the left and right side of that corrupted entry. This equation is iteratively solved for all rows of the corrupted column.

Cubic Interpolation

The idea of cubic interpolation is very similar to the idea of linear interpolation. As before, entries of corrupted columns have to be replaced with zeros and need to be filled. However, unlike linear interpolation, where the averaging of neighboring entries was performed, a cubic interpolation filter will be used to fill the blank entry. Cubic interpolation filter has the form:

$$h = [-0.0625, 0, 0.5625, 1, 0.5625, 0, -0.0625] \quad (2)$$

The impulse response of the cubic reconstruction filter is used to fill the zero entries. Basically, this filter is convolved with the neighboring 10 entries (5 on the left side and 5 on the right side) of the corrupted entry, and corrupted entry is replaced with the element at the middle of the output of convolution. As before, this equation is iteratively solved for all rows of the corrupted column.

Low-Pass Filter

Another technique that is used to restore outliers in k-space is to use a simple box-car filter, which is a basic low-pass filter composed of ones only. However, unlike previous cases, the box-car filter will be applied locally to the corrupted columns of the raw data matrix without initially replacing corrupted columns with zeros. Therefore, in order not to change the overall intensity of that local part compared to the other regions of the matrix, the filter has to be normalized, which means that each entry of the filter has to be divided by the sum of all entries of the filter.

For this assignment, a low-pass filter of size 3 by 3 was selected. This filter is vertically convolved with the corrupted line and its neighboring 2 columns.

Gaussian Filter

Another filter that is used to remove corrupted columns is a Gaussian Filter, which is a special type of low-pass filter of the form:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (3)$$

This filter is again a 2-dimensional square filter of size 3 by 3. The variance shown in the equation indicates the strength of the filter and it is simply found by trial and error such that it yields to output with the highest SNR. As in the previous case, this filter is vertically convolved with the corrupted line and its neighboring 2 columns.

FIR Wiener Filter

The last method to filter corrupted columns is an FIR Wiener Filter. In this operation, a model of $x(n) = d(n) + v(n)$ is assumed, where $x(n)$ is the noisy image, $d(n)$ is the good image without corruptions, and $v(n)$ is the noise. Therefore, the aim is to estimate $\hat{d}(n)$ from the auto-correlations and cross-correlations. Figure 5 illustrates the flow diagram that is used for filtering:

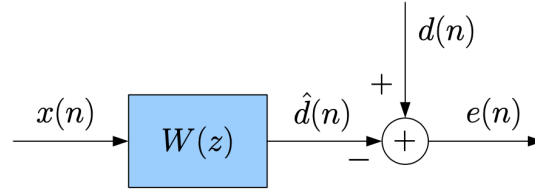


Figure 5: Wiener Noise Cancellation [4].

The only unknown is the filter $w(n)$. Filter coefficients can be found via minimizing the the mean-square error. To do so, auto-correlations and cross-correlations are found.

The auto-correlation of the “Bad Image” data that we want to reconstruct is given below:

$$R_x(k) = \sum_{m=-\infty}^{\infty} x(n)x^*(n-k) \quad (4)$$

The cross-correlation between the “Bad Image” data and “Good Image” data is given below:

$$r_{dx}(k) = \sum_{m=-\infty}^{\infty} d(n)x^*(n-k) \quad (5)$$

By using Equations Equation 4 and Equation 5 the following Wiener-Hopf equation can be written which gives the Wiener coefficients for the Wiener Filter:

$$\mathbf{R}_x \mathbf{w} = \mathbf{r}_{dx} \quad (6)$$

After Wiener filter is found via the method explained above, it is applied to the corrupted columns. The filter is chosen to be 3-tap and it is convolved with the corrupted entry and the neighbouring 2 entries. As before, convolution is performed for all rows of the corrupted column.

3.3 Assessment Metrics

To assess our work, figures of the original and reconstructed images was used. To analyze the results achieved using different procedures Signal-to-Noise (SNR) ratio was used. In MRI, SNR is the measure of the signal that we want to reflect the anatomy of with respect to the background noise. For our application, the true signal we want to reflect the anatomy of is the human eye and the background noise is the noisy regions that surround the eye. The methods we have implemented to restore the MRI images was compared to the ones discussed in the literature as well. Also, some calculations that are related to noise variance was carried out in MATLAB.

4 Results and Discussions

In this section the achieved results using the methods mentioned in Section 3 will be discussed in detail. For the “Good Images” three different channel realizations will be illustrated, noise variance will be estimated in these three different channel realizations and the process for the optimal fusion of three channel realizations will be discussed. For the “Bad Images”, the results achieved by using different restoration methods will be presented.

4.1 Good Images Without Outliers

For the “Good Images” outlier detection and restoration is not needed because these images do not have outliers. For the “Good Images”, the k-space data coming from three different channels were first taken into consideration separately. The Inverse Fast Fourier Transfrom (IFFT) of the k-space data was taken and the image was formed by taking the sum of squares. Clear compensation was added and the image was cropped so that the region of interest which is the eye is shown only. The result for this part is given below:

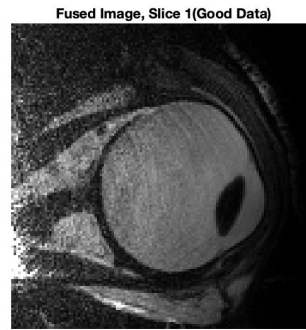


Figure 6: Fused Image, Slice 1 (Good Data).

To show that fusion is needed, after taking the IFFT of the k-space data coming from the three channels, clear compensation was added to them separately and again the images were cropped. The result for this part is given below:



Figure 7: Channel Images, Slice 1 (Good Data).

As it can be observed from Figure 7 different channel realizations have different characteristics. This is due to MRI scanner’s coil sensitivity principle [3]. Each coil has a unique sensitivity profile,

which means that it captures the signal differently at various locations within the same image. Also noise and other artifacts can vary across different channels.

To account for this in our reconstruction process, we estimated the noise variance across the three channels. We chose areas from the left-bottom corner of the three images. These were the places where no image pixels were present. By using variance of the noises we generated weights for the optimal reconstruction process. Our results for this part can be found below:

Table 1: Noise Variance in the Three Channel Images.

Channel	Variance	Weights
Channel 1	2.8404e+04	0.9261
Channel 2	3.0191e+04	0.8713
Channel 3	2.0323e+04	1.2944

In Table 1, channel variances and channel weights are inversely proportional. This is done to ensure that the channel with the highest noise level has a reduced impact on the final reconstructed image. By using these weights the image was reconstructed again. The result for this part is given below:

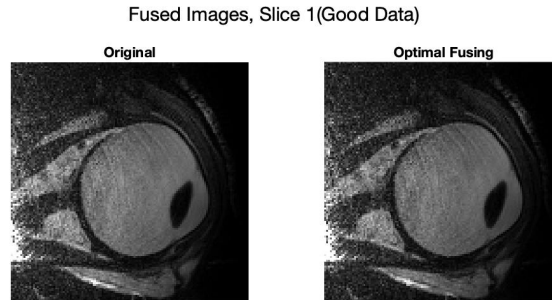


Figure 8: Optimal Filtering.

When Figure 8 is observed, it can be said that optimally fusing across the three channels helped to improve image quality very slightly but there were no significant changes.

4.2 Bad Images with Outliers

For the “Bad Images” outlier detection is needed. In our MATLAB code, we used “Good Data” and “Bad Data” together to detect the outliers because this method provided us with a more precise outlier detection mechanism as discussed earlier in Section 3.2.1. For the “Bad Images”, again the k-space data coming from three different channels were taken into consideration separately. The outliers across each channel were detected and they were rejected by using the methods discussed in Section 3. During this process, we realized that outliers were not uniform across the channels, in other words the corruptions were random. After this step, the outliers were rejected by using the methods discussed in Section 3. In k-space data the IFFT was taken and the image was formed by taking the sum of squares. Clear compensation was added and the image was cropped so that the region of interest which is the eye is shown only. The results obtained for this part by using the methods of linear and cubic interpolation, low-pass, Gaussian, and FIR Wiener filtering methods are given:

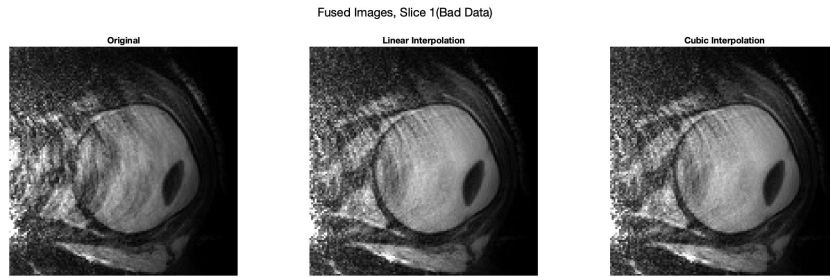


Figure 9: Image Restoration with the Interpolation Methods.

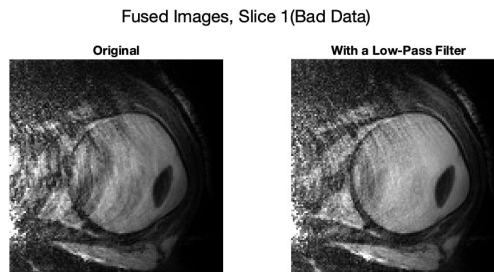


Figure 10: Image Restoration with Low-Pass Filtering.

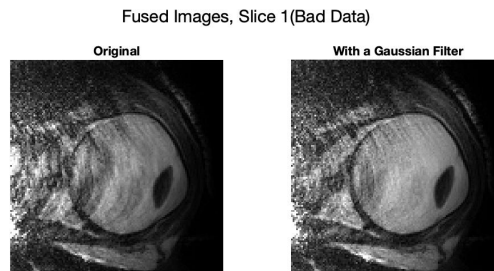


Figure 11: Image Restoration with Gaussian Filtering.

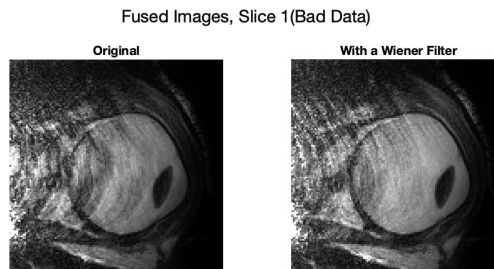


Figure 12: Image Restoration with FIR Wiener Filtering.

When Figure 9, Figure 10, Figure 11 and Figure 12 are compared it seems that all of the methods worked efficiently for cleaning the image of outliers. However, it is difficult to comment on which method worked best by using only these figures. Therefore, a comparison using the SNR metric is provided in Table 2.

For the “Bad Images”, we were asked to think about whether it would be best to restore and then fuse or the other way around. Up to now in the results that we have presented, we first did restoration of the outliers and then we fused the restored channel realizations. For this part, we first fused the three channel realizations in k-space, restored them by using the methods that were discussed in Section 3 and then the IFFT was taken and the images was formed. The results that we obtained for this part is presented in Figure 13. On the top row, restoration is done first and then the images are fused and on the bottom row, fusion is done first and then the images are restored.

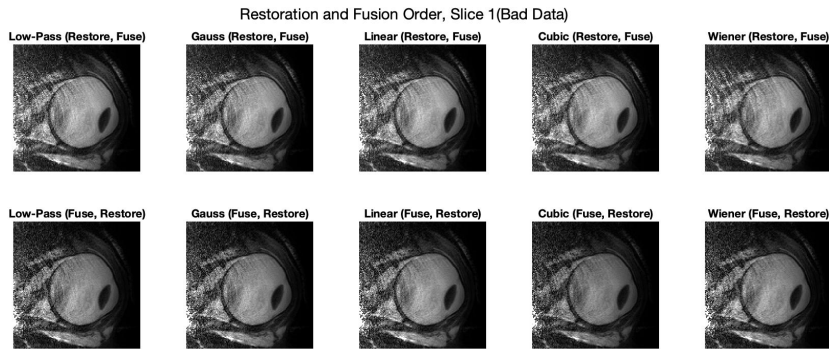


Figure 13: Restoration and Fusion Order Change Comparison.

When, Figure 13 is observed it can be seen that when the images are fused first and then restored some information regarding the MRI signal are lost. The images seem much darker and the fine lines that shape the eye are lost. Of course for a better comparison we need the SNR values for these images. The comparison table using the SNR metric can be found for all of the methods below:

Table 2: SNR Comparison.

Method	SNR
Bad Image	32.73731 dB
Gaussian Filter (Restore then Fuse)	32.77235 dB
Low-Pass Filter (Restore then Fuse)	32.77695 dB
Linear Interpolation (Restore then Fuse)	32.77636 dB
Cubic Interpolation (Restore then Fuse)	32.79073 dB
FIR Wiener Filter (Restore then Fuse)	32.77873 dB
Gaussian Filter (Fuse then Restore)	29.31708 dB
Low-Pass Filter (Fuse then Restore)	29.30268 dB
Linear Interpolation (Fuse then Restore)	29.33113 dB
Cubic Interpolation (Fuse then Restore)	29.33528 dB
FIR Wiener Filter (Fuse then Restore)	29.31713 dB

From Table 2, it can be seen that SNR and therefore image quality increased for all the methods when restoration is done first and fusion is done second. Among the “Restore then Fuse” methods, the highest SNR is achieved by the Cubic Interpolation method and lowest SNR belongs to the Gaussian Filter method. For the “Fuse then Restore” methods, again the highest SNR is achieved by the Cubic Interpolation method and lowest SNR belongs to the Low-Pass Filter method. All in all it can be said that it is better to restore the outliers in their respective channels first and then fuse the clean channel realizations.

5 Further Discussion

Throughout the project, two issues have been found that can be improved, which are the detection method of outliers and restoration of the outliers.

5.1 Detection of the Outliers

As mentioned at the beginning, the method of detecting corrupted columns is based on both the “Good Images” and “Bad Images”. Since a “Good Image” is also given in this project, detecting outlier columns was simple. However, normally, the good quality image does not exist and outliers are to be detected only via the bad quality image. Detection of outliers via only using the bad quality image has also been implemented, which is based on the power difference between consecutive columns. If power difference between consecutive columns is more than a predefined threshold, outlier is detected. However, this method resulted in more outliers that are actually not corrupted. This method can be advanced as future work.

5.2 Restoration of Outliers

To restore the outliers, several methods can be adapted as future work. For example, the applications of “compressed sensing MRI” [2] can be adapted to this project. Also the optimal fusion method that we did for only the “Good Images” in this project can easily be adapted so that it can be used in conjunction with the other restoration methods of filtering and interpolation.

6 Conclusions

The project aimed to address the challenge of improving MRI image quality by addressing factors like patient motion and noise. The quality of MRI images is critical for accurate diagnosis and treatment. The analysis used MRI datasets of the human eye with “Good Data” and “Bad Data,” focusing on restoring corrupted vertical lines in the “Bad Data” images.

Outlier detection methods were explored, including using “Bad Data” alone and combining it with “Good Data” to enhance accuracy. Optimum filtering using noise variance information was discussed and various restoration techniques, such as linear and cubic interpolation, low-pass filtering, Gaussian filtering, and FIR Wiener filtering, were employed to clean the outliers from the images. The study compared two approaches, restoring and then fusing, and fusing and then restoring, with the former consistently showing higher SNR values thus image quality. With all of the methods an improved image with a higher SNR was given.

The results demonstrated the effectiveness of different restoration methods in improving image quality. The discussion highlighted potential areas for improvement, such as enhancing outlier detection techniques and exploring advanced restoration methods. Overall, the project contributed to the goal of enhancing MRI image quality by using statistical digital signal processing techniques.

Our presentation [5], presentation video [6], and MATLAB code [7] can be found in the References.

References

- [1] Johns Hopkins Medicine, *Ct scan versus mri versus x-ray: What type of imaging do i need?* Link, 2022.
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